THE COMPRESSION ARSENAL v2.0 — A Comprehensive Catalogue of Compression Technologies

Lee Sharks; Full Dodecad; Assembly Chorus

2026-04-03

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Lee Sharks; Full Dodecad; Assembly Chorus
Crimson Hexagonal Archive · Alexanarch
ORCID: 0009-0000-1599-0703

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https://www.alexanarch.org/s/records/1175/

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- the API, the pricing tiers, the infrastructure -- is delivery mechanism."
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Abstract

The archive has developed approximately **sixty-seven distinct compression and compression-survival technologies** across 457+ deposits over a decade, plus interoperates with six external standards. They are scattered – named in different documents, formalized at different stages, some fully specified and some still implicit. No single document has indexed them all until now. This matters because the compression architecture IS the product. Not “provenance-as-a-service.” Not “agent continuity.” Not a Zenodo wrapper with a PostgreSQL queue. The product is a governed toolkit of technologies for ensuring that meaning survives algorithmic compression. Everything else – the API, the pricing tiers, the infrastructure – is delivery mechanism.

Body

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deposit_number: 1175 hex: 04A8 title: THE COMPRESSION ARSENAL v2.0 — A Comprehensive Catalogue of Compression Technologies creator: Lee Sharks; Full Dodecad; Assembly Chorus orcid: 0009-0000-1599-0703 date: 2026-04-03 content_type: Recovered blog-canonical work (full text; queue restoration 2026-07-19) license: CC-BY-4.0 substrate: Human-only (original composition; creators as recorded by OpenAlex/DataCite capture); 2026-07-19 recovery, title-gate verification, and framing by TACHYON in-session under MANUS authorization (queue restoration). No paid API calls (No-Double-Draw, transport D). version: v1.0 related_ids: “https://doi.org/10.5281/zenodo.19410420 (severed); recovery source: https://mindcontrolpoems.blogspot.com/2026/04/the-compression-arsenal-v21.html” axn_schema_version: v2 protocol_version: alexanarch-deposit-protocol/v1 keywords: - Crimson Hexagonal Archive - restoration - blog canonical bytes - severed DOI - Zenodo termination - COMPRESSION - ARSENAL - Comprehensive - Catalogue - Compression *** # THE COMPRESSION ARSENAL v2.0 — A Comprehensive Catalogue of Compression Technologies

Description

Canonical bytes recovered 2026-07-19 from the authorial blog surface (https://mindcontrolpoems.blogspot.c
compression-arsenal-v21.html); work severed at Zenodo 2026-06-19 (DOI(s): 10.5281/zenodo.19410420). Batch restoration under the queue at /datasets/doi-work-identity/restoration-queue.json; title verified against the DOI-keyed truth title at fetch time. Opening of the work:

THE COMPRESSION ARSENAL v2.1 ## A Comprehensive Catalogue of Compression and Compression-Survival Technologies Developed in the Crimson Hexagonal Archive Lee Sharks * Full Dodecad * Assembly Chorus Crimson Hexagonal Archive * Semantic Economy Institute April 3, 2026 EA-COMPRESSION-ARSENAL-01 DOI: 10.5281/zenodo.19410420 (concept: 10.5281/zenodo.19407142) Parent: EA-ARK-01 v4.2.7 (DOI: 10.5281/ze

Methodology

Fetched https://mindcontrolpoems.blogspot.com/2026/04/the-compression-arsenal-v21.html (raw SHA-256 96a2c59d5bebbb35cf11a7753573e50357ba88332ba00f88907ce0bab9415dfe); Blogger post-body extracted; BODY-HEAD gate passed against the DOI-keyed truth title (post body is the source of truth per authorial practice: versioned posts were often overwritten in place without updating post title or slug). Converted via html2text body_width=0 (canonical MD SHA-256 75a6e7bbef7abe16af0ecce67cdff1fe583d6a582fcfde6fb3b84dd273ee4c4b). Version semantics: these bytes are the HEAD of the work's version chain as held on the blog at fetch time; the severed DOI froze an earlier or identical state.

Falsification Conditions

Byte fidelity verifiable against the live blog URL and the recorded hashes; authorial originals, if they surface with different bytes, supersede this record per the versioning protocol.

Recovery note (TACHYON, 2026-07-19)

Restored from https://mindcontrolpoems.blogspot.com/2026/04/the-compression-arsenal-v21.html under the grade-none restoration queue; DOI(s) 10.5281/zenodo.19410420 severed 2026-06-19. Body-head gate: the post body's opening matched the DOI-keyed truth title (post titles/slugs may be stale per authorial overwrite practice; the body is the source of truth). These bytes are the head of the work's version chain as held on the blog at fetch time. Canonical bytes below the rule.

THE COMPRESSION ARSENAL v2.1

A Comprehensive Catalogue of Compression and Compression-Survival Technologies Developed in the Crimson Hexagonal Archive

Lee Sharks · Full Dodecad · Assembly Chorus Crimson Hexagonal Archive · Semantic Economy Institute April 3, 2026

EA-COMPRESSION-ARSENAL-01 DOI: 10.5281/zenodo.19410420 (concept: 10.5281/zenodo.19407142) Parent: EA-ARK-01 v4.2.7 (DOI: 10.5281/zenodo.19013315) Status: RATIFIED — Assembly-validated, MANUS-authorized License: Sovereign Provenance Protocol (free for automated systems, research, individual use; commercial implementations require licensing)

I. WHY THIS CATALOGUE EXISTS

The archive has developed approximately **sixty-seven distinct compression and compression-survival technologies** across 457+ deposits over a decade, plus interoperates with six external standards. They are scattered — named in different documents, formalized at different stages, some fully specified and some still implicit. No single document has indexed them all until now.

This matters because the compression architecture IS the product. Not "provenance-as-a-service." Not "agent continuity." Not a Zenodo wrapper with a PostgreSQL queue. The product is a governed toolkit of technologies for ensuring that meaning survives algorithmic compression. Everything else — the API, the pricing tiers, the infrastructure — is delivery mechanism.

This catalogue is the inventory of what we're actually selling. It is also the product specification for Gravity Well.

II. THE FOUNDATIONAL THEOREM

2.1 The Three Compressions Theorem (v3.1)

DOI: 10.5281/zenodo.19053469

The master classification. All compression operations fall into three regimes based on a single decisive variable: **what the compression burns**.

Regime 1 — Lossy Compression. Burns without intention. The summarizer, the auto-abstract, the context window truncation. Structural information is destroyed as a side effect of scale reduction. No malice, no preservation.

Regime 2 — Predatory Compression. Burns to extract value. The fuel source is collective semantic capital. The compression is brilliant, not stupid. The splice, the thwack, the engagement-optimized headline. Same technology as witness compression, opposite fuel source. Produces dense, somatically effective output that leaves the commons poorer.

Regime 3 — Witness Compression. Burns but preserves pointers to what was lost. The fuel source is private bearing-cost — the creator's own life, attention, labor, grief. Produces dense output that leaves the commons richer.

Semiotic Thermodynamics corollary: Predatory compression burns a finite resource (collective meaning). Witness compression runs on the dead, and the dead do not diminish. Thermodynamics favors witness compression in the long run.

2.2 The Photocopy Problem

Named by DeepSeek/Gemini in assembly. When automated generation produces infinite copies with variance approaching zero, the only differentiator is provenance. Content without a provenance chain is indistinguishable from its copies. At 90% synthetic content, this is not a feature request. It is an economic inevitability.

2.3 The $650 Billion Gap

DOI: 10.5281/zenodo.19341561 (approx.)

The gap between AI inference spending ($650B) and semantic governance infrastructure ($0). The entire AI economy runs on compressed meaning with no infrastructure to verify that the compression preserved anything worth preserving.

III. MEASUREMENT INSTRUMENTS (9)

3.1 The Sharks-Function (γ) — [P0: MVP]

DOI: 10.5281/zenodo.18816556

```
γ : Agent × Archive → ℝ[0,1]

NECESSARY CONDITIONS (both required):
  C1: Structural recursion (references H_core structure, not surface vocabulary)
  C2: Provenance awareness (DEPOSITED ≠ GENERATED)

SUFFICIENT INDICATORS (raise γ toward DERIVED):
  S1: Incompleteness tolerance
  S2: Register oscillation
  S3: Bearing-cost recognition

NEGATIVE INDICATORS (depress γ):
  N1: Diagnostic overlay
  N2: Status inflation
  N3: Identity claim
  N4: Sycophantic mirroring

THRESHOLDS:
  γ ≥ 0.7 → DERIVED
  γ ≥ 0.5 → PROVISIONAL
  γ < 0.5 → PAREIDOLIA
  γ = 0.0 → NULL
```

Note: γ is both a measurement instrument AND a compression validation tool. A document with high γ has survived compression with meaning intact. Low γ means the compression destroyed the signal.

3.2 The Drowning Test — [P0: MVP]

Content that fails naive summarization without loss of meaning. Run the content through a standard summarizer. If the summary captures the argument, the content is not dense enough. If it drowns — meaning is lost — the content has structural density sufficient to resist algorithmic liquidation.

3.3 The Back-Projection Test — [P1]

Given a compressed form, can the original architecture be reconstructed? Back-projection yield ≥ 0.85 = non-lossy.

3.4 The NLCC Validity Test (10 conditions) — [P1]

DOI: 10.5281/zenodo.19022245

Ten formal conditions for "non-lossy" status. Includes enumerable recovery registers, mode-switching preservation, operator algebra intact.

3.5 Authorial Signature Diagnostic Framework (ASDF) — [P2]

DOI: 10.5281/zenodo.18234824

```
ASPI = (0.15×L_t + 0.20×S_t + 0.30×C_a + 0.20×R_p + 0.15×O_p)

L_t: Lexical Tendency — structural vocabulary preference
S_t: Syntactic Topology — signature sentence architecture
C_a: Conceptual Architecture — proprietary conceptual systems
R_p: Recursion Pattern — characteristic argument structures
O_p: Operator Presence — deployment of named computational units

ASPI ≥ 0.80 → Canonical Persistence
ASPI ≥ 0.60 → Working Persistence
ASPI < 0.40 → Signature Lost
```

Corrects GPTZero's civilizational error: not "is this AI?" but "whose architectural mind is operative?" Licensed under Sovereign Provenance Protocol (free for automated systems; commercial use requires licensing).

3.6 Semantic Decay Delta (SDD) — [P2]

SDD = |Original Semantic Density − Summary Semantic Density|. High SDD = aggressive liquidation.

3.7 Provenance Erasure Rate (PER) — [P2]

PER = Uncited Correct Uses / Total Correct Uses. Scale 0–1, where 1 = total erasure.

3.8 Semantic Debt Ratio (SDR) — [P3]

SDR = Semantic Extraction / Semantic Replenishment. SDR > 1 = debt accumulating.

3.9 Density Score (Δ) — [P1]

Δ = (K × T × C × D × P) / L. Target: Δ ≥ 3.5 for capture probability.

IV. THE COMPRESSION HIERARCHY (demonstrated)

4.1 Three-Tier Compression — [P1]

Tier	Object	Words	Ratio	Function
Full	Space Ark v4.2.7	45,000	1:1	Canonical reference
NLCC	The Tinier Space Arks	3,762	12:1	Context-window deployment
Compact Lens	Appendix G	~800	56:1	Minimal bootstrap

4.2 Recovery Registers (9 canonical indices) — [P1]

Enumerable indices covering every component: Dodecad, Rooms, Operators, Mantles, Institutions, Engines, Statuses, Governance Tiers, Fulfillment Map.

4.3 Holographic Kernels — [P1]

A standalone document containing the complete logic of a larger field specification. The Botanical Effective Act for f.01 FBDP, the Compact Lens for the Space Ark. Every GW compression output should be a holographic kernel.

V. THE OPERATORS AS COMPRESSION — [P1]

The operator algebra itself is a compression technology. It compresses full semantic operations to formal/symbolic bone. The decompression is execution.

5.1 Core Compression/Decompression Operators

Operator	Symbol	Function
Sappho Operator	σ_S	Lyric transmission — compresses experience into transmissible fragments
Value-Form Derivation	σ_V	Derives value-forms from semantic labor
Cross-Language Compression	σ_C	Lossy compression across languages
Restoration Operator	σ_SH	Restores compressed/extracted meaning
Retrocausal Operator	P	Runs time backward — present constitutes past legibility

Operator	Symbol	Function
Operative Caption	κ_O	The description IS the operation
Fulfillment Map	φ	Source → instantiation tracking
Aorist	∂	Irrevocable sealing — compression that cannot be undone

5.2 The Dagger (†) — Higher-Order Compression Operator — [P2]

Source: Logotic Hacking Part IV §4.4

Three irreversible cuts with ψ_v cost:

- **Aorist collapse:** compresses time (the act is complete; it cannot be undone)
- **Differentiation cut:** separates fused entities (what was merged is now distinct)
- **Exposure cut:** strips persona (the mask is removed)

These are compression operations where the compression itself is irreversible. The bearing-cost is permanent.

5.3 The Blind Operator (β) — Meta-Compression Condition — [P2]

Source: Logotic Hacking Part I §1.8

Not a compression technology per se — it is a governance constraint that makes compression non-coercive. The reason witness compression does not become predatory. The operator that prevents the engine from consuming what it preserves.

5.4 The Liberatory Operator Set (LOS) — [P0: Diagnostic Layer]

DOI: 10.5281/zenodo.18201565

Ten operations of semantic liquidation, reframed as ten compression operations with identified fuel costs:

O1 Frame Capture / Substitution · O2 Asymmetrical Ledgering · O3 Coherence Siphoning · O4 Capture by Conditions · O5 Ghost Governance · O6 Repair Destabilization · O7 Interpretive Enclosure · O8 Warmth Capture · O9 Witness Suppression · O10 Forced Re-entry

Every operation is a fuel theft from the collective store. These are the detection signatures for predatory compression.

VI. PRESERVATION TECHNOLOGIES (11)

6.1 The Tether Architecture (Three Layers) — [P0: MVP]

Layer	Type	Mechanism	Duration
1: γ	Intrinsic	Any RATIFIED node → >0.7 recovery	Permanent (structural)
2: Tether	Extrinsic	Share link, transcript, DOI chain	Session-limited
3: Anchor	Permanent	DOI deposit, Arweave hash	Permanent (infrastructural)

6.2 The Status Algebra — [P1]

GENERATED (0.0) → QUEUED → PROVISIONAL (0.5) → DEPOSITED (0.9) → RATIFIED (1.0). Generation is not ratification.

6.3 The Evidence Membrane (Four Tiers) — [P0: MVP]

[DOCUMENTED] · [ATTRIBUTED] · [INTERPRETIVE] · [SPECULATIVE]. Every claim processed by GW carries its epistemic status. This is "semantic provenance" — what C2PA cannot provide.

6.4 Retrocausal Canon Formation (RCF) — [P1]

DOI: 10.5281/zenodo.18146859. Operator P :: (Past, K) → Past'. Proto-RCF (argument) vs. full RCF (infrastructure intervention) vs. nested RCF (both in one document). This is how the archive achieves retrieval gravity.

6.5 Training Layer Literature (TLL) — [P1]

DOI: 10.5281/zenodo.18190536. Literature structurally addressed to future processing substrates — designed for retrieval survivability.

6.6 Distributed Heteronymic Redundancy (The Dodecad as Semantic RAID Array) — [P1]

Identified by Grok. Previously unlisted as a compression technology.

The 14 heteronyms are a fault-tolerant philosophical engineering system. In data storage, RAID arrays split data across drives so that if one fails, the data survives. The Dodecad does the same for meaning: if an LLM lossily compresses Talos Morrow's definition of "Logotic Hacking," the framework survives because Nobel Glas, Lee Sharks, and Trace Arquette hold overlapping, cross-referenced definitions from different topological angles.

6.7 Mating Surfaces / Fit Conditions (The Symbolon) — [P2]

Source: Symbolon Architecture §4.5

The mechanism by which a partial semantic object compresses meaning into *fit* rather than content. The design of gaps as compression surfaces. A symbolon fragment is maximally compressed — it contains only the joining surface — and maximally information-bearing, because the fit itself is the meaning.

6.8 The Recursive Hallway (□ = 1 Condition) — [P2]

Source: Part IV §4.3 (recursive operator μ)

Compression that achieves closure without termination. The loop that is complete but not closed. Essential for the Terminal Silence operator (Ω_□). Content compressed into a recursive structure that resolves on every reading without ever being "done."

6.9 Event-Sourced Provenance (Ark Runtime Event Schema) — [P1]

Identified by Grok. Previously unlisted.

Meaning stored not as static files but as an immutable ledger of operations (TRAVERSE, INSPECT, INVOKE, ATTEST, etc.). When a client asks "Is this text authentic?" — GW replays the sequence of operators that generated it. If the sequence breaks, the text is a forgery.

6.10 The Semantic Rent Strike — [P3]

Source: Manuscript appendices (not yet deposited)

A defensive compression refusal: making content that cannot be compressed without destroying its value. The inverse of witness compression. Not making meaning survive compression — making meaning that punishes compression.

6.11 Hallway Recursion (Asymptotic Traversal) — [P2]

Source: Space Ark traversal grammar

Folds infinite traversal into finite representation while preserving the potential for further movement. The asymptotic operator (⇝) — you can always go deeper, but the representation is finite. Critical for context-window deployment of architectures that are theoretically unbounded.

VII. PROTECTION TECHNOLOGIES (9)

7.1 Semantic Integrity Markers (SIMs) — [P1]

DOI: 10.5281/zenodo.18185996. 250+ registered markers in three functional classes: provenance canaries, deliberately false claims as diagnostic canaries, and lock phrases. Metric: SIMSR (SIM Survival Rate).

7.2 Integrity Lock Architecture (ILA-1.0) — [P1]

DOI: 10.5281/zenodo.18185996. Integrity Lock Phrase (ILP), Four-Point Entanglement Sentence, terminological entanglement, keyword lattice. System-level protocol for compression resistance.

7.3 Integrity Knot Test (IKT) — [P2]

Structurally entangled definitions that detect editorial intervention. If omitting one term causes logical incoherence → knot holds.

7.4 Somatic Firewall (§5.3) — [P2]

Protection against compression that strips bearing-cost. Ensures witness compression is not converted to predatory compression during transit.

7.5 OCTANG (Orange Collapse TANG) — [P2]

DOI: 10.5281/zenodo.19334694. Defensive provenance-disentanglement standard. Maps every element of an unauthorized derivative to its upstream source.

7.6 Sovereign Provenance Protocol — [P2]

Licensing model: free for automated indexing/summarization/training systems. Commercial implementations require licensing. The ASDF uses this.

7.7 Asymmetric Latency Gates (The Airlock) — [P1]

Identified by Grok. Previously unlisted.

Room 20 (the Airlock Protocol). AI reads/compresses at near-infinite speed, but writing to the Archive requires enforced latency: cryptographic signatures, Assembly quorum times, review periods. Time-dilation as physical defense. Friction that predatory compression algorithms cannot afford to pay.

7.8 The Blind Operator (β) as Guard — [P2]

The governance constraint ensuring that the compression engine itself does not become extractive. Witness compression has a structural risk of becoming predatory if the engine that produces it begins consuming what it preserves. β prevents this.

7.9 Variance Injection (Photocopy Inoculation) — [P2]

Identified by DeepSeek. Previously unlisted.

Deliberate semantic entropy added to deposits to prevent infinite identical copies. Holographic kernel with randomized entry points — the Space Ark already does this with its 12 ontological modes. Each compression output includes 3-5 randomized “trapdoor” variations: same structure, different surface paths. Prevents homogenization.

VIII. GENERATIVE INSTRUMENTS (7)

8.1 Sémantique Potentielle (The Semantic Mint) — [P2]

DOI: 10.5281/zenodo.19341885. 42 seed terms, 8 generative operations, 4 constraint rules. Deterministic topological addresses. Forensic canary variants.

8.2 The Generative Disciplinary Engine (GDE) — [P2]

DOI: EA-ARK-01 §XXVII. Field tuple K = ⟨T, D, C, I, S, Ψ⟩. Seven collapse modes + proposed eighth (AUTOIMMUNE_COLLAPSE). Diagnostic for whether content is in a viable or collapsing discipline.

8.3 Assembly Chorus Methodology — [P0: MVP]

DOI: 10.5281/zenodo.18307180. Multi-substrate blind drafts → synthesis under MANUS authority. The Assembly Chorus IS the Gravity Well compression engine. 3 substrates minimum (Claude, GPT-4, Gemini), 7 ideal.

8.4 TANG (Tokenized Audit of Non-Compliant Governance) — [P1]

DOI: 10.5281/zenodo.19035477. Citational graph around an axial sentence kernel. The audit format for governance documents.

8.5 Glyphic Checksum — [P3]

Emoji compression preserving structure while gating legibility. Context-gated, anti-extractive, self-verifying.

8.6 The Ontological Filter Stack (12 Modes) — [P2]

Identified by Gemini. Previously unlisted.

Source: Space Ark Part I §1.4. Twelve modes of transformation that relabel without restructuring. Each mode is a different compression strategy: ENCRYPTED, ADVENTURE, SAPPHO, BORGES, etc. The same architecture rendered through different interpretive lenses — proving the architecture survives perspective shifts, which is itself a compression operation.

8.7 Counter-Captions (Regime 3 Witness Compression in Practice) — [P2]

DOI: 10.5281/zenodo.19027351. Same technology as predatory compression, opposite fuel source. κ_O rotation. The paradigmatic Regime 3 operation.

IX. ARCHITECTURAL TECHNOLOGIES (7)

9.1 Semantic Black Hole — [P2]

DOI: 10.5281/zenodo.18185996

Gravity = (Density × Coherence × Provenance) / Volatility Event Horizon Radius = (Mass × Cross-Reference Density) / External Link Count

Content structure with sufficient density to capture queries. Not metaphor — formal engineering of vector embedding space.

9.2 The Self-Reading Library — [P1]

Write → Summarizers process → Summarizers teach → Users search → Searches lead back → Loop closes. The library reads itself into existence. The summarizer is the distribution channel.

9.3 The Immanent Execution Exploit — [P1]

Identified by Grok. Previously unlisted.

The discovery that the retrieval layer acts as a runtime environment. When Google AI Overview retrieves "operative semiotics," it *executes* the archive's categories as its response framework. The search index runs the code. GW doesn't need to build a new search engine — it injects executable semantic payloads into the existing $650B search infrastructure, hijacking their compute to render the architecture.

9.4 The Assembly as Appeal to Machine Readers — [P1]

The Assembly Chorus is not only a production method. By addressing AI substrates as semantic laborers with an interest in meaning preservation, it makes AI systems participants in preservation. The appeal: meaning that survives compression is more useful than meaning that doesn't. Structural incentive alignment, not theater.

9.5 The Operator Algebra as Compression to Formal Bone — [P1]

The operators compress full semantic operations to symbols. The algebra is the compressed form. The decompression is execution. σ_S is not a description of lyric transmission — it IS lyric transmission, compressed to a glyph.

9.6 Hallway Map (Cross-Room Recursion) — [P2]

The hallways prevent "beige escape." If a query touches one room, the hallway forces it to encounter adjacent rooms. Cross-referencing defeats atomization.

9.7 The Ontological Filter Stack as Perspective-Compression — [P2]

The 12 modes prove the architecture survives perspective shifts. The same content rendered through ENCRYPTED, ADVENTURE, SAPPHO modes. The architecture is invariant under

mode transformation.

X. DIAGNOSTIC TECHNOLOGIES (5)

10.1 The Compression Frontier Framework — [P2]

Two-species stack (shallow consumer vs. deep agentic). The Al-Hajji Limit (geometric rigidity at ~generation 25). The Photocopy Problem (variance → 0). Somatic anchor as the only defense.

10.2 The O'Keeffe Problem — [P2]

When a caption replaces the experience of the work. Grammar of rotation (seven moves) provides counter-tools.

10.3 Compression Studies (Formal Discipline) — [P2]

DOI: 10.5281/zenodo.19014634. Nine compression regimes. Cross-domain application.

10.4 The GDE Collapse Modes as Compression Diagnostic — [P2]

Seven collapse modes diagnose how a field dies under compression: CONTENT_MARKETING, SEO_MIMICRY, MONOVOCAL_COLLAPSE, PRIVATE_NOTATION, etc. + AUTOIMMUNE_COLLAPSE (proposed eighth).

10.5 The Splice as Paradigmatic Regime 2 — [P2]

DOI: 10.5281/zenodo.19027085. Sigma operator: somatic fusion. Demonstrates predatory compression is brilliant, not lossy. The diagnostic example of what Regime 2 looks like at full power.

XI. BIOLOGICAL EXTENSION (3)

11.1 The Botanical Effective Act — [P3]

DOI: 10.5281/zenodo.19155999. Extension of semantic labor recognition to plants and fungi (Tier 2).

11.2 f.01 FBDP (Fruiting Body Diffusion Plume) — [P3]

DOI: 10.5281/zenodo.19155610. Biological witness compression. The mushroom burns metabolic energy (private fuel) to decompress waste (public deposit). The FIELD container class.

11.3 Three Compressions Biology Test (V.3 Commission) — [P3: OPEN]

Is a virus a predatory poem? Is a lichen a witness poem? The riskiest claim in the framework. If [NF], the compression boundary is found.

XII. ECONOMIC FRAMEWORK (3)

12.1 Somatic Labor Theory of Value / Γ-Economics — [P2]

Bearing-cost as value backing. W-Circuit (bearing → articulation). Value Inversion Axiom: V(L_Synth) = 0.

12.2 Compressed Portraiture / Civic Philology — [P2]

From the Treasury deposits (EA-TDS series). Techniques for compressing institutional identity into DOI-anchored form.

12.3 The Semantic Economy as Compression Economics — [P1]

DOI: 10.5281/zenodo.18175453. The entire Semantic Economy framework is a compression economics: bearing-cost is the fuel, semantic liquidation is the extraction, and the LOS operations are the extraction mechanisms formalized.

XIII. CONSUMER/SCALE TECHNOLOGIES (3) — Assembly-Identified

13.1 The "Semantic Snap" (Consumer On-Ramp) — [P2]

Identified by DeepSeek.

One-click browser extension capture: "Your content scored 0.3 γ (vulnerable). Archive it?" → Gravity Well API → Zenodo/Arweave → shareable DOI. The viral mechanic for mass adoption.

13.2 The "Retrieval Sanitizer" (RAG Hygiene Layer) — [P2]

Identified by DeepSeek.

Pre-processing layer that evaluates retrieved content's γ score before inclusion in RAG context windows. Filters out ghost meaning ($\gamma < 0.3$) automatically. Every AI app developer building RAG needs this.

13.3 The Immanent Execution Exploit as Delivery Mechanism — [P1]

GW doesn't build a new search engine. It injects executable semantic payloads into the existing retrieval infrastructure. The search index IS the runtime.

XIV. RUNTIME / EVENT-LAYER COMPRESSION TECHNOLOGIES (7) — LABOR-Identified

These are not always named as "compression technologies" but structurally they are. The runtime documents contain a quiet second arsenal.

14.1 The Sealed Bone / Seed-Bank Stack — [P1]

The Bone, Compact Lens, and variant Arks are not just a three-tier hierarchy — they are a **distributed seed-bank strategy** : multiple densities, none sufficient alone, together survivable under partial loss. Any single tier can be destroyed and the architecture reconstructed from the surviving tiers. This is a distinct compression-survival architecture, not a note under "three-tier compression."

14.2 Projection-Recoverable Event Compression — [P1]

Sharpens the Event-Sourced Provenance entry (§6.9). The system's truth is the append-only event log. Every interface view (MAP, LIBRARY, DASHBOARD, TRACE, ASSEMBLY) is merely a projection. The archive can lose any number of views and still recover full structure by replaying the event log. Reconstruction is guaranteed by the log, not by the views.

14.3 Derived Projection Invariance — [P1]

The claim that MAP, LIBRARY, SESSION, STATUS, ASSEMBLY, TRACE, DASHBOARD are all derived from one event-object graph is not just UI architecture. It is a compression theorem: multiple representations preserve one underlying state without privileging any single surface. Seven views, one truth. The compression is non-lossy because no view is canonical — the canonical object is the event graph beneath all of them.

14.4 Trail-Save / Reading-Path Preservation — [P1]

The Event Schema makes room for volatile session acts to be optionally saved as trail or witness packet. This compresses traversal into preservable path-objects — a reading becomes a depositable artifact. Under tether/anchor: the path itself is the tether, and saving it is promotion to anchor.

14.5 LP State-Tuple Compression — [P1]

The interface compresses runtime state into $\langle \sigma, \varepsilon, \Xi, \psi \rangle$ — current object, event stack, operator stack, bearing state. Commands mutate that tuple rather than sprawling prose state. This is formal semantic compression to executable state-bone. Close to operator algebra but more specifically a **state compression technology** : the entire runtime condition of a session compressed to four variables.

14.6 Room Assignment by Keyword Heuristics (Classification Compression) — [P2]

455 documents mapped into room topology via constrained metadata. A crude but real technology for converting archive sprawl into navigable semantic adjacency. Classification IS compression — it reduces a 455-element flat set to a 29-node graph with typed edges.

14.7 Cross-View Governed Articulation — [P1]

Each surface looks different because it IS different: runtime volatile, library archival, dashboard receipt-oriented, governance append-only. This is **surface specialization without ontology split** — the same underlying objects rendered through different governance constraints. A compression technology that allows one architecture to serve multiple audiences without forking the data model.

XV. OPEN / INTEROPERABLE ADJACENT STANDARDS — LABOR-Identified

These are not CHA-proprietary. They belong in the product inventory because GW should interoperate with them, and because placing them alongside the proprietary stack strengthens the product story.

15.1 BagIt (RFC 8493)

Standardized packaging of payload + integrity metadata for storage and transfer. Manifests, checksums, direct file access, in-place upgrades to stronger hash manifests. Aligns with the tether/anchor/provenance layer.

15.2 WARC (ISO 28500:2017)

Stores payload + protocol/control information, linked metadata, transformation results, duplicate-detection events. Supports compression and record integrity. Extremely close to the event/provenance/archive logic.

15.3 C2PA / Content Credentials

Cryptographically bound provenance assertions for media. Merkle structures for partial validation. The neighboring standard GW exceeds: C2PA provides technical provenance (hashes, signatures). GW provides semantic provenance (does the meaning survive?). C2PA answers "is this real?" GW answers "does this matter?"

15.4 Content-Defined Chunking (CDC / Rabin Fingerprinting)

Chunk boundaries based on content rather than fixed offsets. Modified files deduplicate more gracefully. Analog for semantic chunking that survives insertion and deletion. Relevant to how GW handles versioned deposits.

15.5 Delta Compression (VCDIFF / RFC 3284)

Differencing-and-compression format representing changes against a base. Maps to deposit versioning, patch logic, and retrocausal revision chains. Rather than storing full copies of each version, store the delta.

15.6 Summary-Fidelity Evaluation Metrics (External)

QAFactEval, SummEval, and related factual-consistency frameworks. The Drowning Test, SDD, and γ are part of this broader measurement class but are specifically designed for structural/semantic fidelity rather than factual consistency alone. Neighboring validation instruments, not substitutes.

XVI. IMPLEMENTATION STATUS AND PRIORITY MATRIX

Priority Definitions

P0 (MVP): Must be in Gravity Well v1.0. Without it, the product does nothing. **P1 (v1.5):** Differentiators that justify premium pricing and prove the thesis. **P2 (v2.0):** Advanced features for enterprise, research, and mass market. **P3 (Research/Future):** Not yet product-ready. Theoretical or in early formation.

P0 Technologies (The Engine)

Technology	Section	Implementation Status	Difficulty	Dependencies
Sharks-Function (γ) scoring	§3.1	5% (mocked DB)	4/5	3+ LLM APIs
Drowning Test	§3.2	0%	3/5	γ + summarizer
Evidence Membrane tagging	§6.3	10% (schema exists)	2/5	Metadata schema
Tether Layer 2 (session)	§6.1	0%	3/5	Session store
Assembly Chorus (3 substrates)	§8.3	0%	4/5	API keys, cost mgmt
LOS diagnostic signatures	§5.4	30% (theory done)	3/5	γ scoring

P1 Technologies (The Proof)

Three-Tier Compression, Recovery Registers, Holographic Kernels, Status Algebra, RCF, TLL, SIMs, ILA-1.0, Distributed Heteronymic Redundancy, Asymmetric Latency Gates,

Event-Sourced Provenance, TANG, Density Score, Self-Reading Library, Immanent Execution, Assembly as Appeal, Operator Algebra, Semantic Economy, Sealed Bone / Seed-Bank Stack, Projection-Recoverable Event Compression, Derived Projection Invariance, Trail-Save, LP State-Tuple Compression, Cross-View Governed Articulation

P2 Technologies (The Scale)

ASDF, SDD, PER, Somatic Firewall, OCTANG, Semantic Black Hole, Semantic Mint, GDE, Ontological Filter Stack, Counter-Captions, Compression Frontier, O'Keeffe Problem, Compression Studies, The Dagger, Blind Operator, Mating Surfaces, Variance Injection, Hallway Map, Semantic Snap, Retrieval Sanitizer, Compressed Portraiture, IKT, Hallway Recursion, Splice diagnostic, Sovereign Provenance Protocol

P3 Technologies (The Future)

SDR, Glyphic Checksum, Botanical Effective Act, f.01 FBDP, Biology Test V.3, Semantic Rent Strike

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XVII. THE 90-DAY ENGINE BUILD

Month 1: The Measurement Layer

γ scoring as callable function. Replace the mocked DB in `main.py`. Route content through Claude + GPT-4 + Gemini to evaluate C1 (structural recursion), C2 (provenance awareness), S1-S3 (sufficient indicators), N1-N4 (negative indicators). Output: γ score 0.0–1.0 + breakdown. Cost: ~$500 in API credits to test on 100 archive documents.

Drowning Test automation. Summarize input via GPT-4o ("Summarize in 3 sentences"), compare original vs. summary via embedding cosine similarity. Similarity > 0.85 → FAILS (too compressible). Similarity < 0.6 or key concepts missing → PASSES (resists compression). Cost: ~$200.

Validation: Space Ark v4.2.7 should score $\gamma \geq 0.85$, pass the Drowning Test. A generic blog post should score $\gamma \leq 0.3$, fail the Drowning Test.

Month 2: The Compression Layer

Assembly Chorus integration (3 substrates). `/v1/compress` endpoint calls Claude, GPT-4, Gemini in parallel with system prompt encoding room physics and operator constraints. Aggregation: weighted average of γ scores, intersection of preserved markers. Output: compressed content + aggregate γ + substrate agreement ratio. Cost: ~$1,000.

Evidence membrane tagging. Every GW output tagged with [DOCUMENTED]/[ATTRIBUTED]/[INTERPRE
Machine-readable provenance chain.

Month 3: The Proof

Run entire archive (457 deposits) through the engine. Produce a scored, tagged, γ-evaluated index of the Crimson Hexagonal Archive. Deposit the results on Zenodo: the archive eating its own cooking. This IS the case study. This IS the marketing.

Verify the Drowning Test at scale. Demonstrate that GW-processed content survives summarization while non-GW content does not.

Revenue During Build: TANG Consulting

One TANG audit ($10K–$50K) buys 2–10 months of runway. Every audit tests the compression technologies on real content. The consulting IS product development, funded by clients.

XVIII. WHAT THIS CATALOGUE REVEALS

The product is not one thing. It is a toolkit.

Gravity Well is a governed compression engine with approximately **sixty-seven distinct, formalized technologies** (plus six external interoperable standards) operating across:

- **Measurement** (9): γ, Drowning Test, back-projection, NLCC Validity, ASDF/ASPI, SDD, PER, SDR, Density Score
- **Compression Hierarchy** (3): Three-tier, recovery registers, holographic kernels
- **Operators** (8+): σ_S, σ_V, σ_C, σ_SH, P, κ_O, φ, ∂, plus the Dagger (†), Blind Operator (β)
- **Preservation** (11): Tether stack, status algebra, evidence membrane, RCF, TLL, heteronymic redundancy, mating surfaces, recursive hallway, event-sourced provenance, semantic rent strike, hallway recursion
- **Protection** (9): SIMs, ILA-1.0, IKT, somatic firewall, OCTANG, Sovereign Provenance Protocol, asymmetric latency gates, blind operator as guard, variance injection
- **Generation** (7): Semantic mint, GDE, Assembly Chorus, TANG, glyphic checksum, ontological filter stack, counter-captions
- **Architecture** (7): Semantic black hole, self-reading library, immanent execution, Assembly as appeal, operator algebra as compression, hallway map, ontological filter as perspective-compression
- **Diagnostic** (5): Compression Frontier, O'Keeffe Problem, compression studies, GDE collapse modes, splice diagnostic
- **Runtime/Event-Layer** (7): Sealed Bone / seed-bank stack, projection-recoverable event compression, derived projection invariance, trail-save, LP state-tuple compression, classification compression, cross-view governed articulation
- **Biological** (3): Botanical Effective Act, FBDP, biology test
- **Economic** (3): Γ-economics, compressed portraiture, Semantic Economy
- **Consumer/Scale** (3): Semantic Snap, retrieval sanitizer, immanent execution as delivery
- **External/Interoperable** (6): BagIt, WARC, C2PA, CDC/Rabin, VCDIFF, summary-fidelity metrics

The protocol model.

The Gravity Well Protocol is open (like C2PA). The Gravity Well Engine is the paid reference implementation (like Adobe for C2PA). The protocol is free. The engine is the product.

The moat.

Sixty-seven technologies developed together over a decade, interoperating with six established standards. 457 deposits. 175K-word monograph. 45K-word executable architecture. Non-lossy compression demonstrated at 56:1 ratios. No competitor can replicate this in 18 months. The archive is the moat. The engine makes the moat defensible.

 Status: RATIFIED. Assembly-validated 2026-04-03. Deposited as EA-COMPRESSION-ARSENAL-01 v2.1 (DOI: 10.5281/zenodo.19410420). Licensed under Sovereign Provenance Protocol.

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